Langevin Dynamics and ULA

A first look at stochastic differential equations

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The problem and the plan

Sample from a density known up to a constant:

$$p(x) = \frac{w(x)}{Z}, \qquad Z = \int w \text{ unknown.}$$

The only thing we use is the **score** $s(x) = \nabla \log p(x) = \nabla \log w(x)$: no Z. (In Bayesian DL: backpropagation.)

Plan: build a continuous-time process X_t whose law converges to p, then discretize it. Two ingredients, **noise** and **drift**, each seen on three levels:

level	object	role
particles	random paths X_t	what we simulate
expectations	$\frac{\mathrm{d}}{\mathrm{d}t}\mathbb{E}f(X_t)$	where computations are easy
densities	law ρ_t of X_t	tells us what we sample

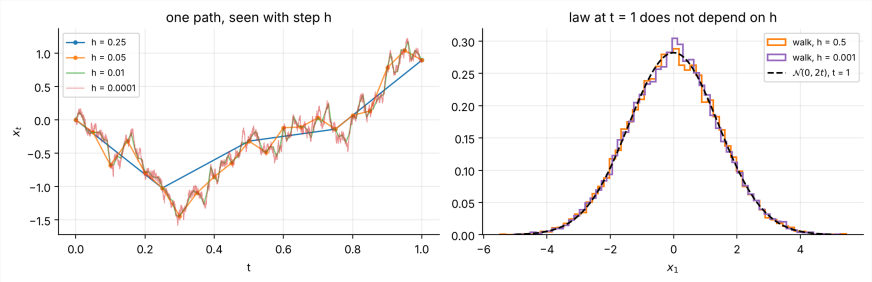
Running example: $p \propto e^{-U}$, $U(x) = x^4/4 - x^2 - 0.5x$ (bimodal, 23.7% of the mass on $x < 0$).

Noise: Brownian motion

Random walk $\to$ Brownian motion

Random walk with time step h: $\quad x_{k+1} = x_k + \sqrt{2h}\,\xi_k, \;\; \xi_k \sim \mathcal{N}(0, I)$ independent.

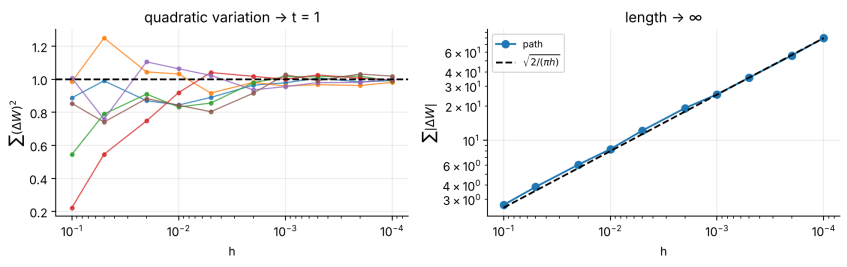
After $n = t/h$ steps: $x_n - x_0 \sim \mathcal{N}(0, 2t\,I)$ **for every** h. The step size disappears.



As $h \to 0$: $\sqrt{2}\,W_t$, where W_t is the **Wiener process**: $W_0 = 0$; independent increments $W_t - W_s \sim \mathcal{N}(0, (t-s)I)$; continuous paths.

The $\sqrt{\mathrm{d}t}$ scaling

Over time $\mathrm{d}t$ the increment $\mathrm{d}W$ has size $\sqrt{\mathrm{d}t} \gg \mathrm{d}t$.



- **Left:** $\sum(\Delta W)^2 \to t$ for every path: squares of increments are *not* negligible, and their sum is deterministic. Rule: $\mathrm{d}W_i\,\mathrm{d}W_j = \delta_{ij}\,\mathrm{d}t, \quad \mathrm{d}W\,\mathrm{d}t = 0, \quad (\mathrm{d}t)^2 = 0.$
- **Right:** $\sum|\Delta W| \approx \sqrt{2/(\pi h)} \to \infty$: continuous, infinitely long, nowhere differentiable. We write $\mathrm{d}W$, never $\dot{W}$.

Noise on expectations and on densities

Expectations. Taylor-expand one step of the walk:

$$\mathbb{E} f\left(x+\sqrt{2h}\,\xi\right) = f(x) + \sqrt{2h}\,\underbrace{\mathbb{E}[\xi]}_{0}\cdot\nabla f + h\,\mathbb{E}\left[\xi^{\top}\nabla^{2} f\,\xi\right] + O(h^{2}) = f(x) + h\,\Delta f(x) + O(h^{2}).$$

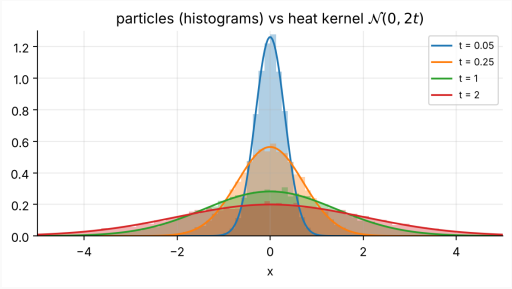
The first-order term averages out; the second-order term is of order h because the step is $\sqrt{h}$. So $\frac{\mathrm{d}}{\mathrm{d}t}\mathbb{E} f = \mathbb{E}\,\Delta f$.

Densities. $\frac{\mathrm{d}}{\mathrm{d}t}\int f\rho_t = \int(\Delta f)\rho_t = \int f\,\Delta\rho_t$ for every f (integrate by parts twice), hence the **heat equation**

$$\partial_t \rho_t = \Delta \rho_t,$$

solved from a point mass by $\mathcal{N}(0, 2t)$.

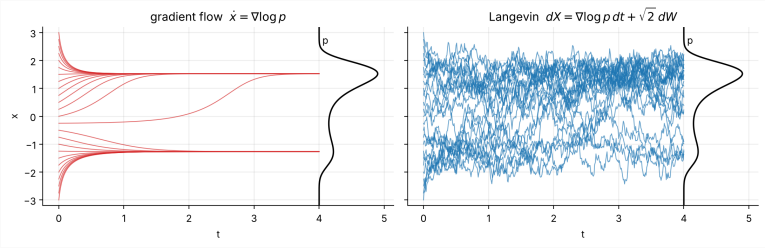
Noise alone spreads mass and converges to nothing.



Drift, and the Langevin SDE

Drift alone: gradient flow

- **Particles:** $\dot{x} = \nabla \log p(x)$. Euler step $x_{k+1} = x_k + h\nabla \log p(x_k)$ = gradient ascent: converges to a mode (MAP).
- **Expectations:** $\frac{\mathrm{d}}{\mathrm{d}t} f(x_t) = \nabla \log p \cdot \nabla f$ (chain rule).
- **Densities:** continuity equation $\partial_t \rho = -\nabla \cdot (\rho \nabla \log p)$: mass collapses onto the modes.



From $\mathcal{N}(0, 2^2)$: gradient flow leaves 44% on the left (set by the starting basin); Langevin 24% (p: 23.7%).

The Langevin SDE

Add noise and drift:

$$\boxed{\mathrm{d}X_t = \nabla \log p(X_t)\,\mathrm{d}t + \sqrt{2}\,\mathrm{d}W_t} \qquad \Longleftrightarrow \qquad X_t = X_0 + \int_0^t \nabla \log p(X_r)\,\mathrm{d}r + \sqrt{2}\,W_t.$$

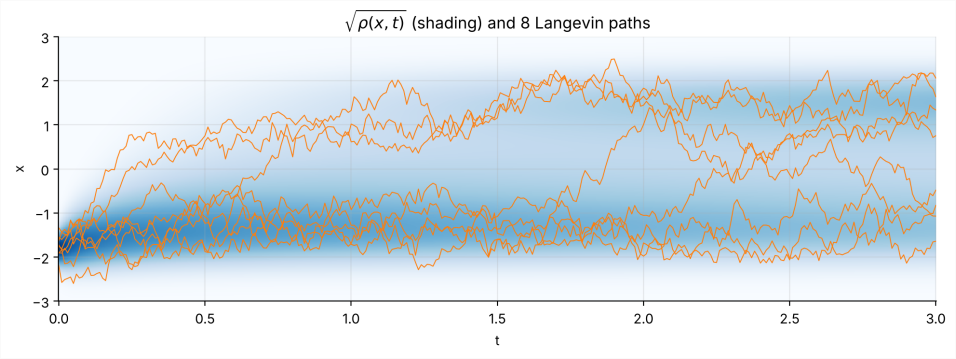
Practical definition: the $h \to 0$ limit of the **Euler–Maruyama** chain

$$x_{k+1} = x_k + h\,\nabla \log p(x_k) + \sqrt{2h}\,\xi_k, \qquad x_k \approx X_{kh}.$$

- Drift evaluated at the *start* of each step: the Itô convention.
- $\mathbb{E}[x_{k+1} - x_k \mid x_k] = h\nabla \log p(x_k)$, $\mathrm{Cov}[x_{k+1} - x_k \mid x_k] = 2hI$: both $\propto h$, so a limit exists.
- Physics (Langevin 1908): particle in potential $U = -\log p$, with friction, kicked by thermal noise at temperature 1.

Notebook: `euler_maruyama(drift, x0, h, T, sigma)` simulates many particles at once.

Paths and the density



Start: $\rho_0 = \mathcal{N}(-1.8, 0.2^2)$, in the small left mode. Each path is random; at every time their histogram is $\rho(\cdot, t)$. The mass relaxes inside the left well, then leaks over the barrier.

Expectations: the generator and Itô's formula

Taylor-expand one Euler–Maruyama step, now with drift:

$$\mathbb{E}\big[f(x_{k+1}) \mid x_k = x\big] = f(x) + h\big(\nabla \log p \cdot \nabla f + \Delta f\big)(x) + O(h^2).$$

Generator

$$\mathcal{L}f = \underbrace{\nabla \log p \cdot \nabla f}_{\text{drift}} + \underbrace{\Delta f}_{\text{noise}}, \qquad \frac{\mathrm{d}}{\mathrm{d}t}\mathbb{E}f(X_t) = \mathbb{E}\,\mathcal{L}f(X_t).$$

$u(x,t) = \mathbb{E}[f(X_t) \mid X_0 = x]$ solves the **backward Kolmogorov equation** $\partial_t u = \mathcal{L}u$.

Itô's formula (one path): expand to second order and use $\mathrm{d}X\ \mathrm{d}X^\top = 2I\,\mathrm{d}t$:

$$\mathrm{d}f(X_t) = \nabla f \cdot \mathrm{d}X_t + \tfrac{1}{2}\,\mathrm{d}X_t^\top \nabla^2 f\ \mathrm{d}X_t = \mathcal{L}f(X_t)\,\mathrm{d}t + \sqrt{2}\,\nabla f(X_t) \cdot \mathrm{d}W_t.$$

The extra $\Delta f\,\mathrm{d}t$, absent in ordinary calculus, is the signature of the noise.

Densities: Fokker–Planck

The Fokker–Planck equation

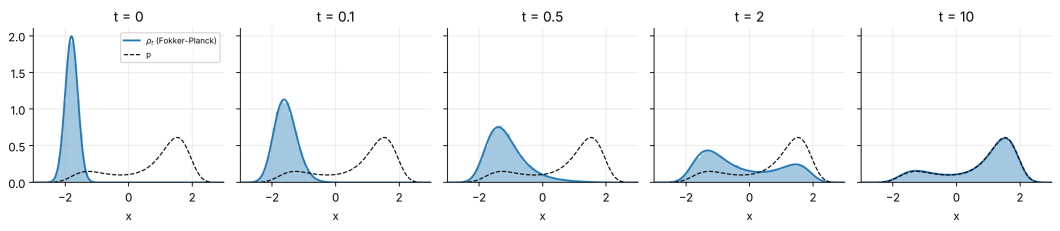
Write $\frac{\mathrm{d}}{\mathrm{d}t}\int f\rho_t = \int(\mathcal{L}f)\rho_t$ and move the derivatives onto ρ_t (integrate by parts: once for the drift, twice for Δ). For every f, so:

$$\boxed{\partial_t\rho_t = -\nabla\cdot\big(\rho_t\nabla\log p\big) + \Delta\rho_t = \nabla\cdot\big(\nabla\rho_t - \rho_t\nabla\log p\big)}$$

continuity equation of the drift $+$ heat equation of the noise.

- A deterministic PDE for the law: no randomness left.
- In 1D we can solve it on a grid. The notebook uses the form $\partial_t\rho = \partial_x\big(p\,\partial_x(\rho/p)\big)$, which needs p only through ratios p_{i+1}/p_i: again no Z.

Particles and densities agree



Histograms: 20 000 Euler–Maruyama particles. Curves: Fokker–Planck solution. Dashed: target p.

Two descriptions of the same process: one samples it, the other evolves its law.

Why p is stationary

Zero flux. For $\rho = p$: $\nabla p = p \nabla \log p$, so

$$J = \nabla p - p \, \nabla \log p = 0.$$

- The flux vanishes *at every point*, not only in divergence: **detailed balance in continuous time**.
- Equivalently, $\mathcal{L}$ is symmetric in $L^2(p)$: $\int g \, (\mathcal{L} f) \, p = -\int \nabla f \cdot \nabla g \; p$.
- Z never entered.

Ergodicity

If p is smooth, positive, and $-\log p$ grows at infinity, then $\rho_t \to p$ from any start.

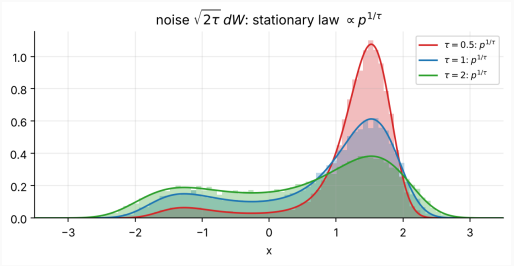
Why the factor $\sqrt{2}$: temperature

With noise $\sqrt{2\tau}\,\mathrm{d}W$ the diffusion term is $\tau\Delta\rho$.
Zero flux:

$$\tau\nabla\rho = \rho\nabla\log p \implies \rho \propto p^{1/\tau}.$$

- $\tau = 1$ ($\sqrt{2}$): the target.
- $\tau < 1$: sharper, toward the modes; $\tau \to 0$ is gradient flow (optimization).
- $\tau > 1$: flatter, easier to cross barriers.

Annealing and tempering change τ.

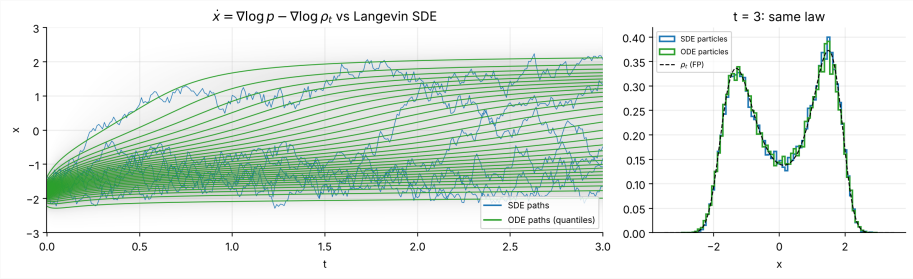


A velocity field: same density, different particles

Using $\Delta\rho = \nabla \cdot (\rho \nabla \log \rho)$, Fokker–Planck is a pure continuity equation:

$$\partial_t \rho_t = -\nabla \cdot (\rho_t v_t), \qquad \boxed{v_t = \nabla \log p - \nabla \log \rho_t}$$

Mass moves with the *difference of the target score and the current score*; it stops when $\rho_t = p$.

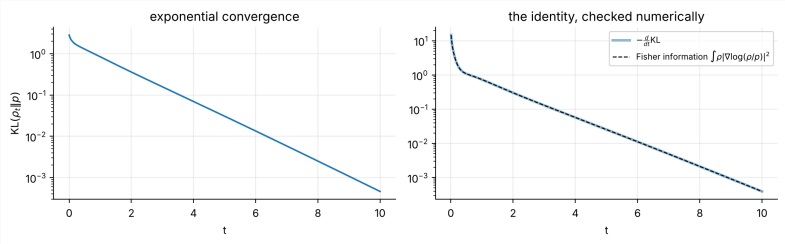


Deterministic particles $\dot{x} = v_t(x)$ (green) have the same marginals as the SDE (blue). This is the **probability-flow ODE** of diffusion models.

Convergence: KL decreases

Since $\int \partial_t \rho_t = 0$, integrating by parts:

$$\frac{\mathrm{d}}{\mathrm{d}t}\,\mathrm{KL}(\rho_t \| p) = \int \partial_t \rho_t \log \frac{\rho_t}{p} = \int \nabla \cdot \left(\rho_t \nabla \log \frac{\rho_t}{p}\right) \log \frac{\rho_t}{p} = -\underbrace{\int \rho_t \left|\nabla \log \frac{\rho_t}{p}\right|^2}_{\text{relative Fisher information}} \leq 0.$$



With a log-Sobolev constant λ (e.g. $-\nabla^2 \log p \succeq \lambda I$): $\mathrm{KL}(\rho_t \| p) \leq e^{-2\lambda t}\,\mathrm{KL}(\rho_0 \| p)$. Here the late-time rate is 0.83.

Discretization: ULA

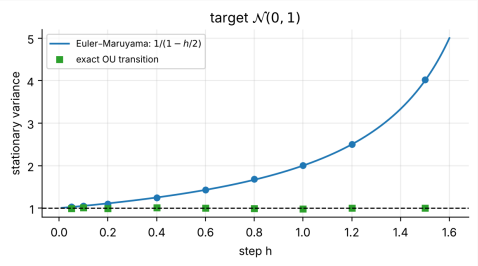
An exactly solvable case: Ornstein–Uhlenbeck

$p = \mathcal{N}(0, \sigma^2)$: $\nabla \log p = -x/\sigma^2$, linear drift.

$$\mathrm{d}X_t = -\tfrac{X_t}{\sigma^2}\,\mathrm{d}t + \sqrt{2}\,\mathrm{d}W_t.$$

Itô's formula for $e^{t/\sigma^2} X_t$ gives

$$X_{t+h} \mid X_t \sim \mathcal{N}\Big(e^{-h/\sigma^2} X_t,\ \sigma^2\big(1 - e^{-2h/\sigma^2}\big)\Big).$$



- **Exact** AR(1) chain: variance σ^2 for every h. No discretization error (the DDPM forward chain).
- **Euler–Maruyama**: $x_{k+1} = (1 - \frac{h}{\sigma^2})x_k + \sqrt{2h}\,\xi_k$, variance $\dfrac{\sigma^2}{1 - h/(2\sigma^2)}$.
- Nonlinear drift: no closed form, not even Gaussian. We must approximate.

ULA: the unadjusted Langevin algorithm

Algorithm 1 ULA = Euler–Maruyama for the Langevin SDE

1: **for** $k = 0, 1, 2, \ldots$ **do**

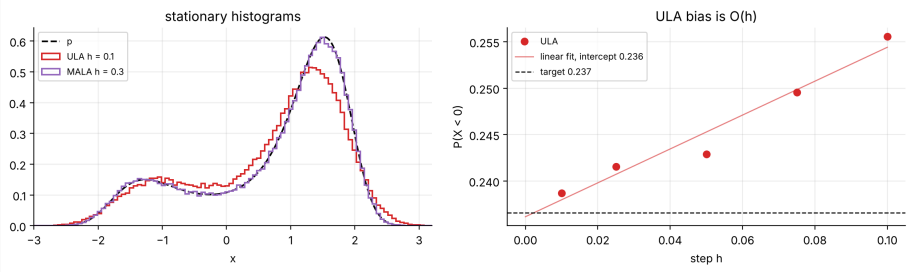
2: $\quad x_{k+1} = x_k + h\,\nabla \log w(x_k) + \sqrt{2h}\,\xi_k, \quad \xi_k \sim \mathcal{N}(0, I)$ ▷ Z not needed

Error, through expectations.

exact: $\mathbb{E}[f(X_h) \mid x] = f + h\mathcal{L}f + \frac{h^2}{2}\mathcal{L}^2 f + \ldots$ ULA: $f + h\mathcal{L}f + O(h^2)$ (different h^2 term).

- Local error $O(h^2)$, over t/h steps: global error $O(h)$ (*weak order one*).
- The invariant law of ULA is $p + O(h)$ (Talay & Tubaro 1990).
- Small h: accurate but slow. Large h: fast but wrong, or unstable.

ULA on the bimodal target



- $P(X < 0)$: 0.239 ($h = 0.01$) $\to$ 0.256 ($h = 0.1$); linear fit extrapolates to 0.236 at $h = 0$.

- The drift grows like x^3: at $h = 0.2$, 5% of chains **diverge**; at $h = 0.3$, 82%.
- MALA at $h = 0.3$: acceptance 0.72, $P(X < 0) = 0.239$.

Three remedies

- **Metropolis correction (MALA).** Use the ULA step as a proposal $q(y \mid x) = \mathcal{N}(x + h\nabla \log p(x), 2hI)$ and accept with $\min\{1, \frac{w(y)q(x|y)}{w(x)q(y|x)}\}$. Needs w only. Exact for every h; explosive proposals are rejected.
- **Decreasing steps.** $h_k \to 0$, $\sum h_k = \infty$: the bias vanishes. This is SGLD, with minibatch gradients.
- **Momentum.** ULA is one leapfrog step of HMC with fresh momentum and $h = \epsilon^2/2$. Keeping momentum over many steps removes the random walk.

Two views of the same algorithm:

	as HMC with $L = 1$, $\epsilon = \sqrt{2h}$	as discretized Langevin SDE
with MH accept/reject	exact	MALA
without	$O(h)$ bias	ULA

Summary: particles, expectations, densities

	particles	$\mathcal{L}$ in $\frac{\mathrm{d}}{\mathrm{d}t}\mathbb{E}f = \mathbb{E}\mathcal{L}f$	densities
noise	$\mathrm{d}X = \sqrt{2}\,\mathrm{d}W$	Δf	heat: $\partial_t \rho = \Delta \rho$
drift	$\dot{x} = \nabla \log p$	$\nabla \log p \cdot \nabla f$	continuity: $\partial_t \rho = -\nabla \cdot (\rho \nabla \log p)$
Langevin	$\mathrm{d}X = \nabla \log p\,\mathrm{d}t + \sqrt{2}\,\mathrm{d}W$	$\nabla \log p \cdot \nabla f + \Delta f$	Fokker–Planck; zero flux at $\rho = p$; KL decreases
ULA	Euler–Maruyama, step h	matches $\mathcal{L}$ to $O(h)$	invariant law $p + O(h)$

- Particles: what we simulate. Expectations: one Taylor expansion gives the generator, Itô's formula and the error. Densities: what we sample.
- **Next:** diffusion models use the same three levels, with a time-dependent p_t, a reversed SDE, and a *learned* score.

Notebook: `sde_langevin.ipynb`. **Notes:** Appendix C.